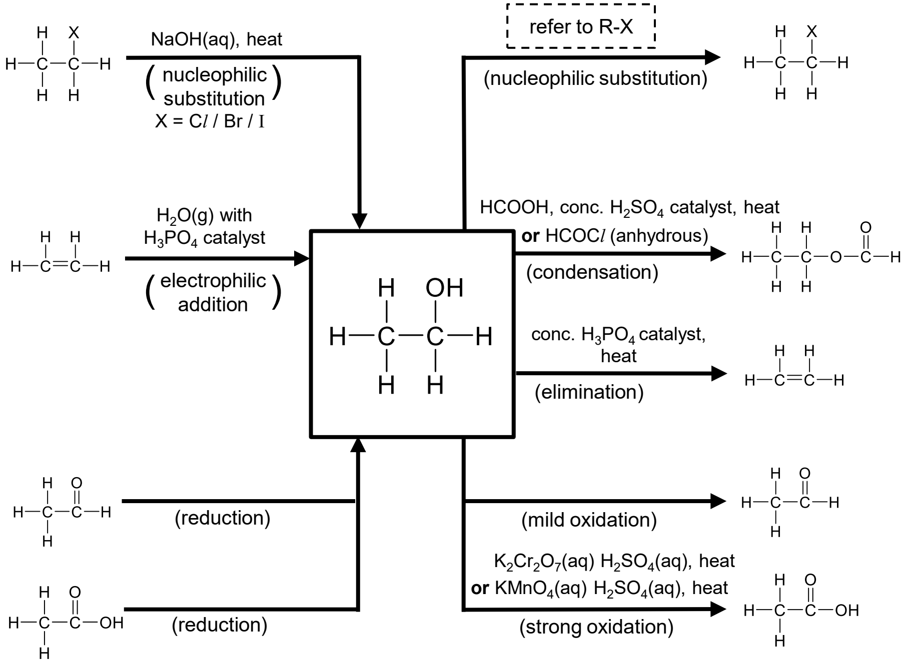
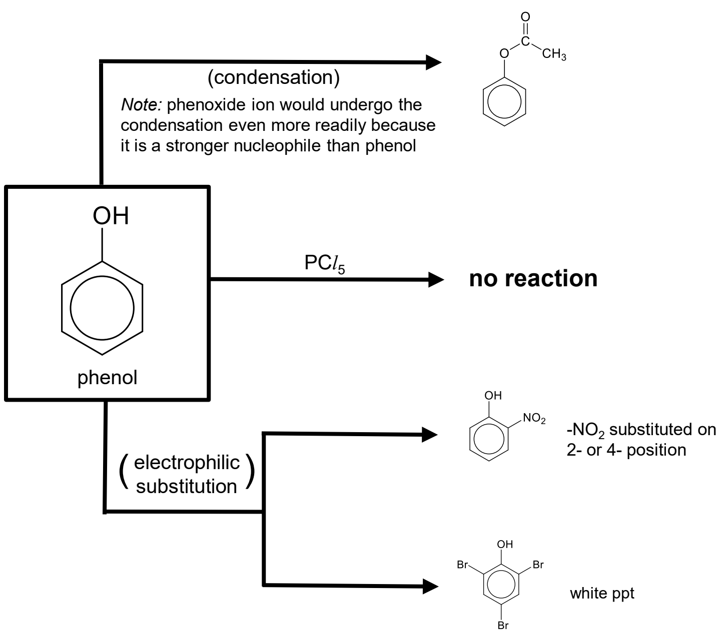
Reactions





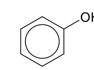
Oxidation

Relative acidity

Key concept: The greater the ________________ of the conjugate base, the __________________ the acid.

➢ Draw the structures of the conjugate bases, and hence rank the following compounds in terms of their acid strength.

acid strength			
compounds	H_2O	CH_3CH_2OH	$CH_3-C(CH_3)_2-OH$
conjugate base		$CH_3CH_2O^-$	
stability of conjugate base			

acid strength			
compounds	CH_3CH_2OH		$CH_3C(=O)-OH$
conjugate base	$CH_3CH_2O^-$		
stability of conjugate base			

Note:

- electron-donating alkyl groups ________________ the negative charge on O^- in the conjugate base, destabilising the ion

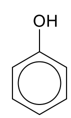
Note:

The phenoxide ion is relatively stable because…

- p orbital of O^- overlaps with π e^- cloud of benzene ring
- lone pair of e^- on O^- ________________ into benzene ring
- ____________ negative charge on O, hence stabilising the ion

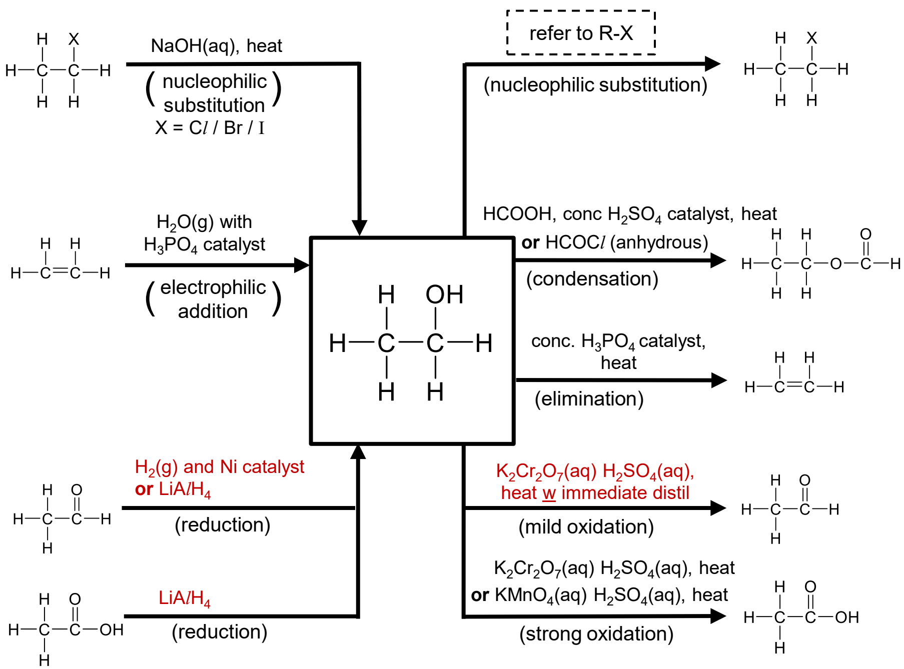
Distinguishing tests

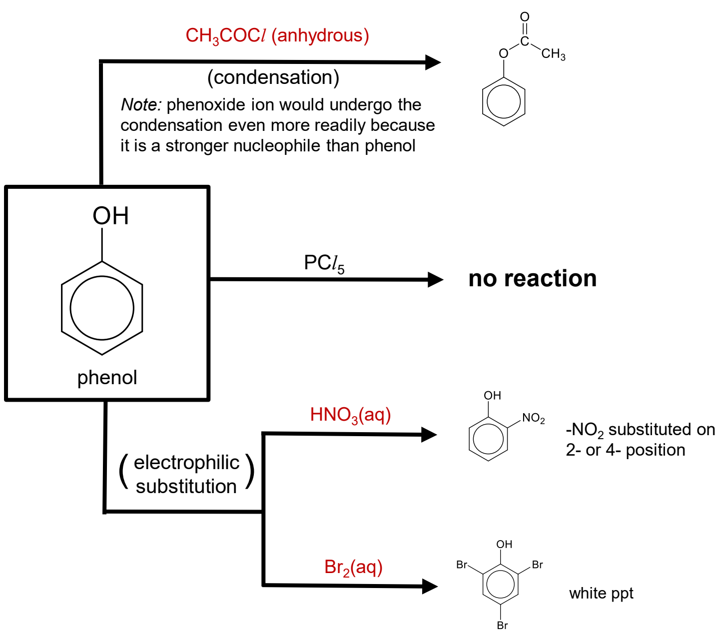
➢ Label which pair of reactants reacts. State any products formed.

	Na	NaOH	Na_2CO_3
CH_3CH_2OH alcohol			
 phenol			
$CH_3C(=O)-OH$ carboxylic acid			

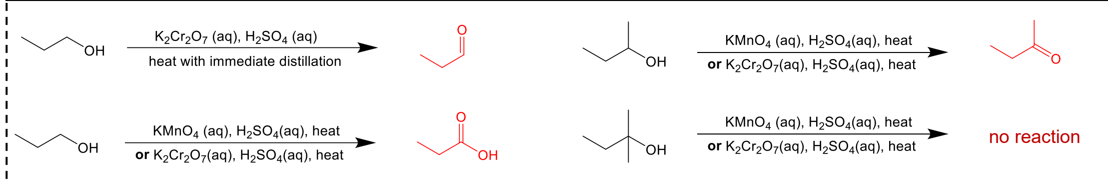
structure	reagents	products & observation
$CH_3-CH(OH)-R$ R = H/ C	I_2 (aq), NaOH (aq), warm	
$CH_3-C(=O)-R$ R = H/ C		

Reactions





Oxidation

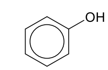
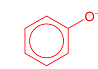


Relative acidity

Key concept: The greater the __stability__ of the conjugate base, the __stronger__ the acid.

➢ Draw the structures of the conjugate bases, and hence rank the following compounds in terms of their acid strength.

acid strength	strongest	→	weakest
compounds	H_2O	CH_3CH_2OH	$(CH_3)_3C{-}OH$
conjugate base	OH^-	$CH_3CH_2O^-$	$(CH_3)_3C{-}O^-$
stability of conjugate base	most stable	→	least stable

acid strength	weakest	→	strongest
compounds	CH_3CH_2OH		$CH_3C(=O){-}OH$
conjugate base	$CH_3CH_2O^-$		$CH_3C(=O){-}O^-$
stability of conjugate base	least stable	→	most stable

Note:

- electron-donating alkyl groups __intensifies__ the negative charge on O^- in the conjugate base, destabilising the ion

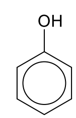
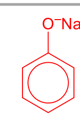
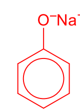
Note:

The phenoxide ion is relatively stable because…

- p orbital of O^- overlaps with π e^- cloud of benzene ring
- lone pair of e^- on O^- __delocalises__ into benzene ring
- __dispersing__ negative charge on O, hence stabilising the ion

Distinguishing tests

➢ Label which pair of reactants reacts. State any products formed.

	Na	NaOH	Na_2CO_3
CH_3CH_2OH alcohol	$CH_3CH_2O^-Na^+$ + $^1/_2$ H_2 (g)	X	X
 phenol	 + $^1/_2$ H_2(g)		X
$CH_3C(=O){-}OH$ carboxylic acid	$CH_3C(=O){-}O^-Na^+$ + $^1/_2$ H_2(g)	$CH_3C(=O){-}O^-Na^+$	$CH_3C(=O){-}O^-Na^+$ + CO_2 (g) + H_2O

structure	reagents	products & observation	
$CH_3{-}CH(OH){-}R$, R = H/ C	I_2 (aq), NaOH (aq), warm	CHI_3 yellow ppt	$Na^+\,{}^-O{-}C(=O){-}R$
$CH_3{-}C(=O){-}R$, R = H/ C		CHI_3 yellow ppt	$Na^+\,{}^-O{-}C(=O){-}R$